Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The major product formed when 1, 1, 1- trichloro - propane is treated with aqueous potassium hydroxide is:

(A) Propyne

(B) 1- Propanol

(C) 2- Propanol

(D) Propionic acid

Q.2

Which of the following order regarding bond enthalpy $\varepsilon(C-X)$ in an alkyl halide (RX) is correct?

(A) $\varepsilon(\mathrm{C}-\mathrm{Cl})<\varepsilon(\mathrm{C}-\mathrm{Br})<\varepsilon(\mathrm{C}-\mathrm{I})$

(B) $\varepsilon(\mathrm{C}-\mathrm{Cl})>\varepsilon(\mathrm{C}-\mathrm{Br})>\varepsilon(\mathrm{C}-\mathrm{I})$

(C) $\varepsilon(\mathrm{C}-\mathrm{Cl})>\varepsilon(\mathrm{C}-\mathrm{Br})<\varepsilon(\mathrm{C}-\mathrm{I})$

(D) $\varepsilon(\mathrm{C}-\mathrm{Cl})<\varepsilon(\mathrm{C}-\mathrm{Br})>\varepsilon(\mathrm{C}-\mathrm{I})$

Q.3

When toluene is treated with Cl_2 at high temperature without the use of any catalyst, the product obtain is

(A) *o*-chlorotoluene

(B) *m*-chlorotoluene

(C) a mixture of orhto- and para-chlorotoluenes

(D) benzylchloride

Q.4

Which of the following orders regarding the boiling points of the given alkyl chloride is correct?

(A) $\mathrm{CH_3\left(CH_2\right)_3Cl}>\mathrm{CH_3CH_2CHClCH_3}>\mathrm{\left(CH_3\right)_3CCl}$

(B) $\mathrm{CH_3\left(CH_2\right)_3Cl}<\mathrm{CH_3CH_2CHClCH_3}<\mathrm{\left(CH_3\right)_3CCl}$

(C) $\mathrm{CH_3\left(CH_2\right)_3Cl}>\mathrm{CH_3CH_2CHClCH_3}<\mathrm{\left(CH_3\right)_3CCl}$

(D) $\mathrm{CH_3\left(CH_3\right)_3Cl}<\mathrm{CH_3CH_2CHClCH_3}<\mathrm{\left(CH_3\right)_3CCl}$

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Q.5

Which of the following reagents is most effective in replacing $-OH$ group of an alcohol by $-Cl$ group?

(A) HCl gas

(B) Concentrated HCl

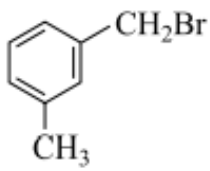
(C) $SOCl$

(D) $SOCl_2$,

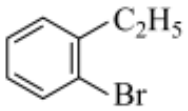
Q.6

Compound (A) C_8H_9Br, gives a white precipitate when warmed with alcoholic $AgNO_3$. Oxidation of (A) gives an acid (B), $C_8H_6O_4$. (B) easily forms anhydride on heating. Identify the compound (A).

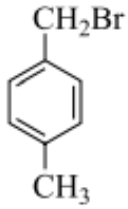
(A)



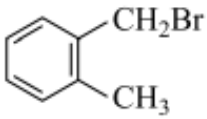
(B)



(C)

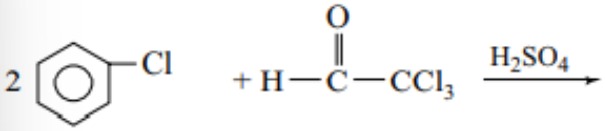


(D)



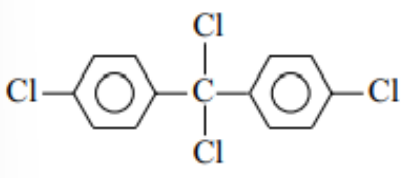
Q.7

Chlorobenzene reacts with trichloro acetaldehyde in the presence of H_2SO_4



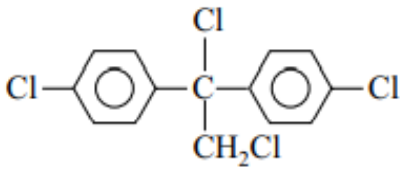
The major product formed is:

(A)

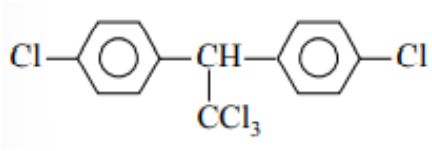


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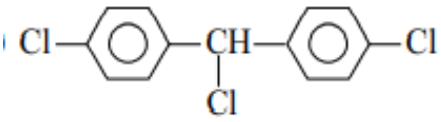
(B)



(C)



(D)



Q.8

In a nucleophilic substitution reaction: $\mathrm{R-Br+Cl^- \overset{DMF}{\rightarrow} R-Cl+Br^-}$, which one of the following undergoes complete inversion of configuration?

(A) $\mathrm{C_6H_5CHC_6H_5Br}$

(B) $\mathrm{C_6H_5CH_2Br}$

(C) $\mathrm{C_6H_5CHCH_3Br}$

(D) $\mathrm{C_6H_5CCH_3C_6H_5Br}$

Q.9

Reaction of trans $2-$phenyl$-1-$bromocyclopentane on reaction with alcoholic KOH produces

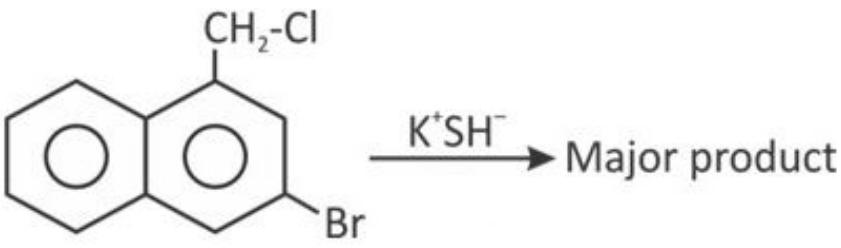
(A) $1-$phenylcyclopentene

(B) $3-$phenylcyclopentene

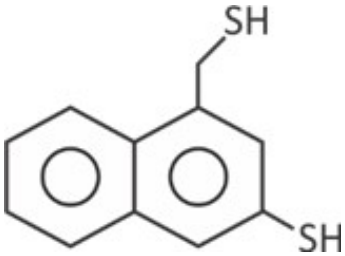
(C) $4-$phenylcyclopentene

(D) $2-$phenylcyclopentene

Q.10

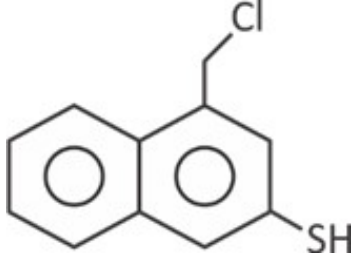


(A)

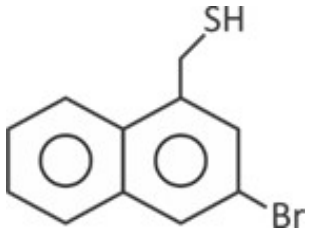


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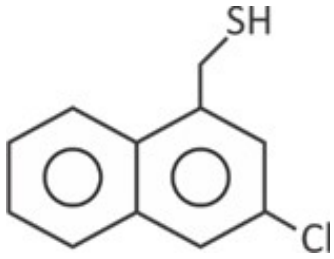
(B)



(C)



(D)



Q.11

The major product in the dehydrohalogenation of 3-Bromo-2, 2-dimethylbutane is

(A) 3, 3-dimethyl but-1-ene

(B) 2, 3-dimethylbut-1-ene

(C) 2, 3-dimethyl but-2-ene

(D) 4-methylpent-2-ene

Q.12

Complete the missing links

$$CH_3CHBrCH_3 \xrightarrow{alc.KOH} X \xrightarrow{HBr,Peroxiden} Y \xrightarrow{CH_3ONa} Z$$

(A) **X** **Y** **Z**

$$CH_3CH = CH_2 \quad CH_3CH(Br)CH_2Br \quad CH_3CH(OH)CH_3$$

(B) **X** **Y** **Z**

$$CH_3CH = CH_2 \quad CH_3CH_2CH_2Br \quad CH_3CH_2OCH_2CH_3$$

(C) **X** **Y** **Z**

$$CH_3CH = CH_2 \quad CH_3CH(Br)CH_3 \quad CH_3CH_2OCH_2CH_3$$

(D) **X** **Y** **Z**

$$CH_3CH = CH_2 \quad CH_3CH_2CH_2Br \quad CH_3CH_2CH_2OCH_3$$

Q.13

Which of the following alkyl halides is hydrolysed by S_N1 mechanism?

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(A) CH_3Cl

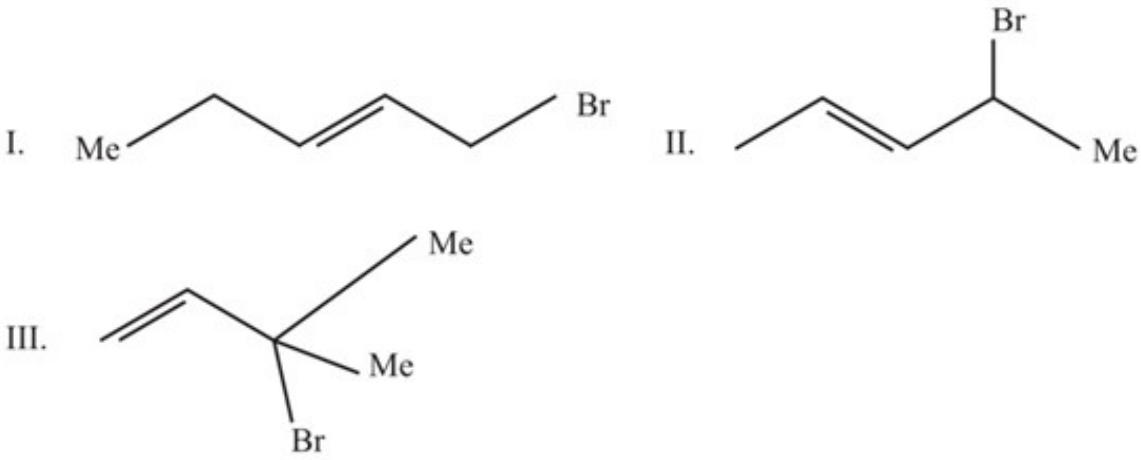
(B) CH_3CH_2Cl

(C) $CH_3CH_2CH_2Cl$

(D) $(CH_3)_3CCl$

Q.14

Which of the following halides will be most reactive in S_N1 reaction and S_N2 reactions, respectively



(A) (I), (II)

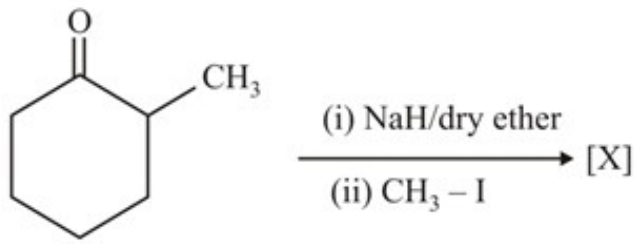
(B) (II), (I)

(C) (I), (III)

(D) (III), (I)

Q.15

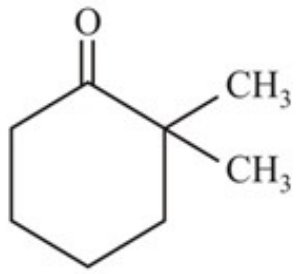
In the given reaction



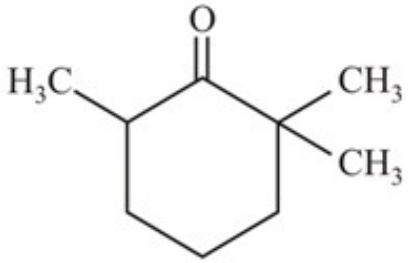
[X] will be

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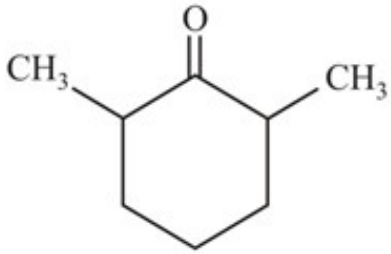
(A)



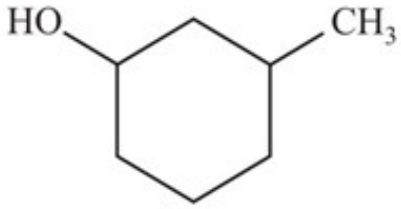
(B)



(C)



(D)



Q.16

(Z)-3-bromo-3-hexene when treated with CH_3O^- in CH_3OH gives

(A) 3-hexyne

(B) 2-hexyne

(C) 2, 3-hexadiene

(D) 2, 4-hexadiene

Q.17

Calculate the percentage of all the monochlorinated products obtained from the chlorination of 2 - methyl butane. The relative reactivity of 1°, 2°, and 3° hydrogen to chlorination is (1 : 3.8 : 5).

(A) is obtained from the reaction of three equivalent $1^{\circ}H$ atoms.

(B) is obtained from the reaction of two equivalent $2^{\circ}H$ atoms.

(C) is obtained from the reaction of one equivalent $3^{\circ}H$ atoms.

(D) is obtained from the reaction of six equivalent $1^{\circ}H$ atoms.

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(A) 15%, 45%, 20%, 25%

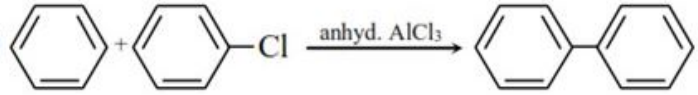
(B) 14%, 35%, 23%, 28%

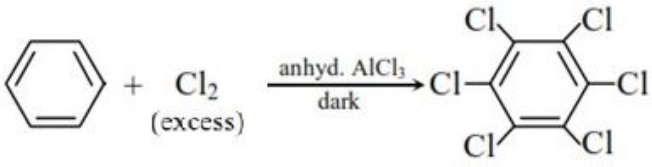
(C) 12%, 30%, 22%, 29%

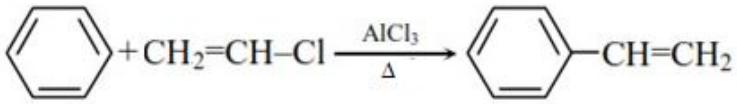
(D) 17%, 36%, 20%, 23%

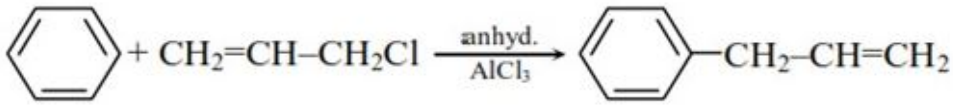
Q.18

Consider the following reaction

(a) 

(b) 

(c) 

(d) 

Which of these reactions are possible?

(A) (b) , (c) and (d)

(B) (b) and (d)

(C) (a) and (d)

(D) (a) and (b)

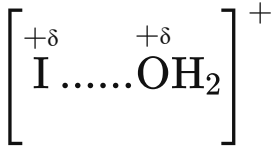
Q.19

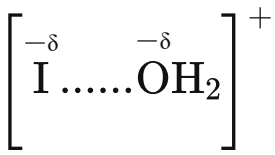
Benzene cannot be iodinated with I_2 directly. However, in presence of oxidants such as HNO_3 , iodination is possible. The electrophiles formed in the case is

(A) $[I^+]$

(B) $[I^-]$

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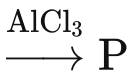
(C) 

(D) 

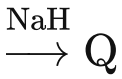
Q.20

Among P, Q, R, S the aromatic compound is

(A) 

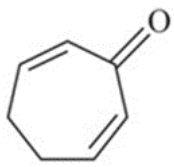


(B) 



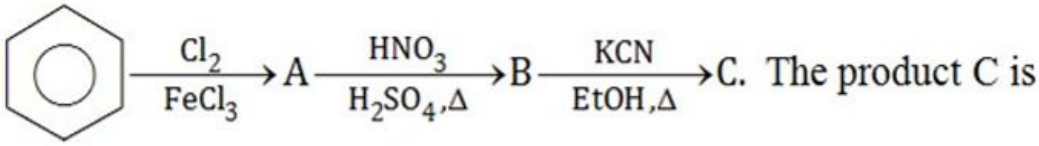
(C) 



(D) 



Q.21

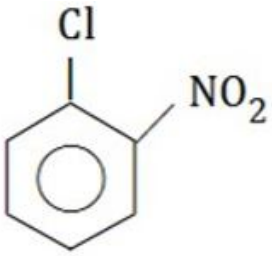


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(A)



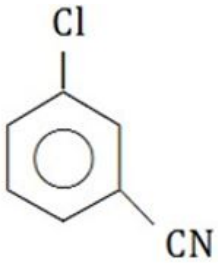
(B)



(C)



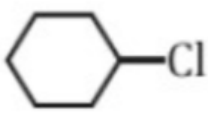
(D)



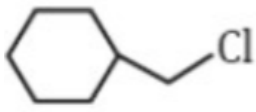
Q.22

Which of the following statement is correct for the reactivity in S_N2 reaction?

(A)



is more reactive than



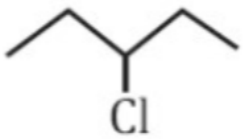
(B)



is more reactive than



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(C) 

is more reactive than

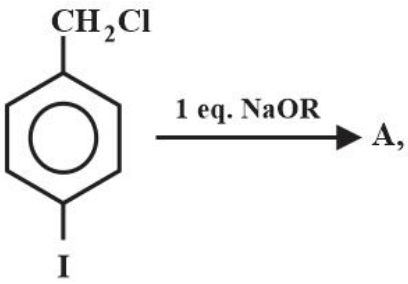


(D) 

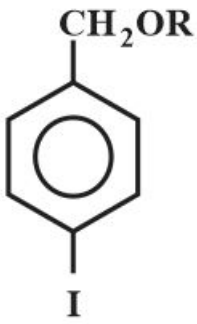
is more reactive than

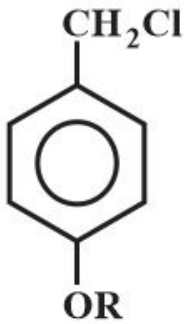


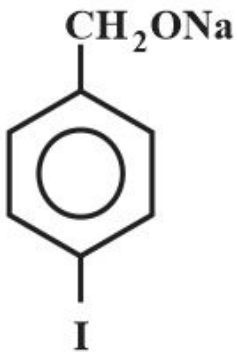
Q.23



A is

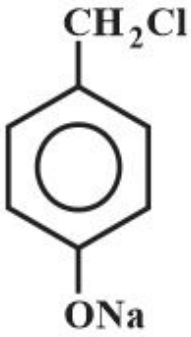
(A) 

(B) 

(C) 

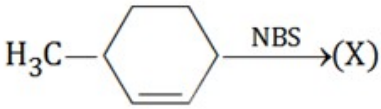
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(D)

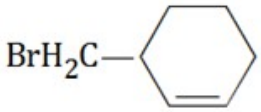


Q.24

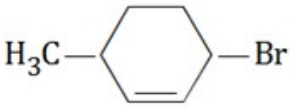
The major product (X) of the monobromination reaction is



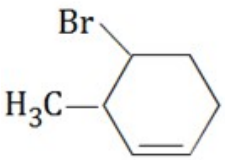
(A)



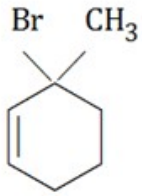
(B)



(C)



(D)



Q.25

The reaction of toluene with Cl_2 in presence of $FeCl_3$ gives ' X' and reaction in presence of light gives ' Y '. Thus, ' X ' and ' Y ' are :

(A) $X =$ Benzal chloride, $Y = o-$ Chlorotoluene

(B) $X = m-$ Chlorotoluene, $Y = p-$ Chlorotoluene

(C) $X = O-$ and $p-$ Chlorotoluene, $Y =$ Trichloromethylbenzene

(D) $X =$ Benzyl chloride, $Y = m$ - Chlorotoluene

Answers & Solutions

Q.1 Answer:

Propionic acid

Solution:

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$$CH_3CH_2CCl_3 + 3KOH \rightarrow [CH_3CH_2C(OH)_3]^+ 3KCl \xrightarrow{-H_2O} \underset{\text{Propionic acid}}{CH_3CH_2COOH}$$

Q.2 Answer:

$\varepsilon(C-Cl) > \varepsilon(C-Br) > \varepsilon(C-I)$

Solution:

Bond enthalpy $C-X$ (where X is Cl, Br, I) decreases with increase in atomic number of X

Q.3 Answer:

benzylchloride

Solution:

At high temperature, substitution at methyl group takes place.

Q.4 Answer:

$CH_3\,(CH_2)_3\,Cl > CH_3CH_2CHClCH_3 > (CH_3)_3\,CCl$

Solution:

Boiling point decreases with increasing branching of the alkyl group.

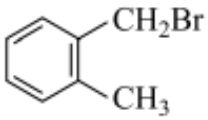
Q.5 Answer:

$SOCl_2$,

Solution:

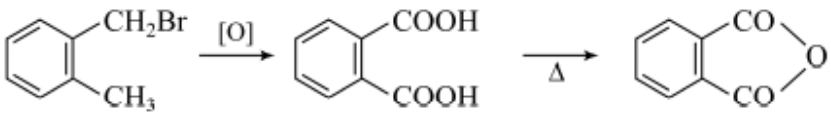
The most effective reagent is $SOCl_2$.

Q.6 Answer:

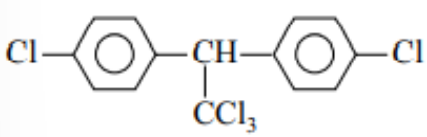


Solution:

Aliphatic Br gives precipitate with $AgNO_3$.The Oxidation of $-CH_2Br$ gives $-COOH$ and also $-CH_3$ gives $-COOH$. The two $-COOH$ group must be in ortho positions to give anhydride. Hence,

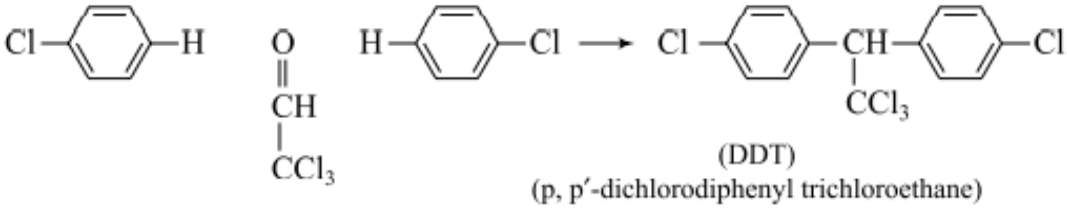


Q.7 Answer:



Solution:

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(DDT)
(p, p′-dichlorodiphenyl trichloroethane)

Q.8 Answer:

$C_6H_5CH_2Br$

Solution:

The reaction involving inversion of configuration proceeds via S_N2 mechanism. The reactivity order is

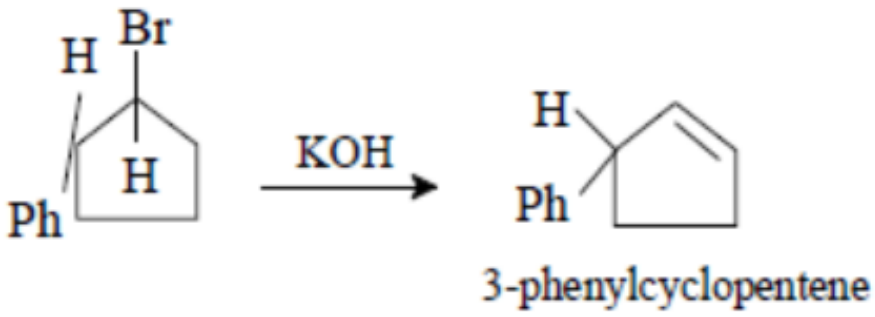
$$CH_3 > 1^\circ > 2^\circ \gg \text{neopentyl} > 3^\circ$$

$C_6H_5CH_2Br$ contains $1^\circ C$ to which Br is attached and will undergo S_N2 mechanism with complete inversion of configuration.

Q.9 Answer:

3—phenylcyclopentene

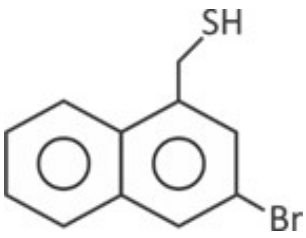
Solution:



3-phenylcyclopentene

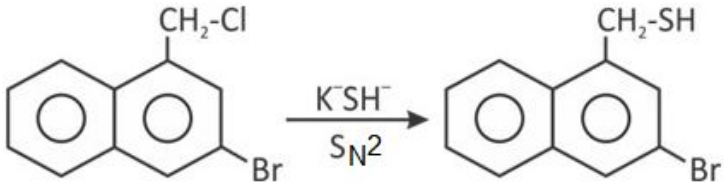
It follows E2 mechanism. Hughes and Ingold proposed that bimolecular elimination reactions take place when the two groups to be eliminated are trans and lie in one plane with the two carbon atoms to which they are attached i.e. E2 reactions are stereoselectively trans.

Q.10 Answer:



Solution:

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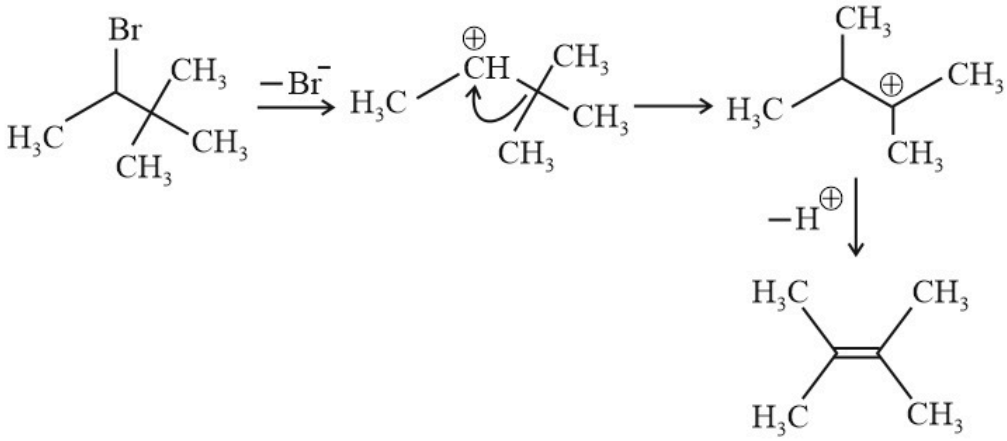


$-Cl$ is the good leaving group in the compound and SH^- is the best nucleophile. So, S_N2 reaction is more favoured.

Q.11 Answer:

2, 3-dimethyl but-2-ene

Solution:



Q.12 Answer:

X	Y	Z
$CH_3CH=CH_2$	$CH_3CH_2CH_2Br$	$CH_3CH_2CH_2OCH_3$

Solution:

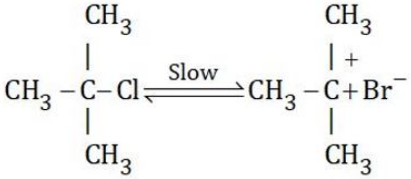
$$CH_3CHBrCH_3 \xrightarrow{\text{alc. KOH}} \underset{(X)}{CH_3CH=CH_2} \xrightarrow{\text{HBr, peroxide}} \underset{(Y)}{CH_3CH_2CH_2Br} \xrightarrow{CH_3ONa} \underset{(Z)}{CH_3CH_2CH_2OCH_3}$$

Q.13 Answer:

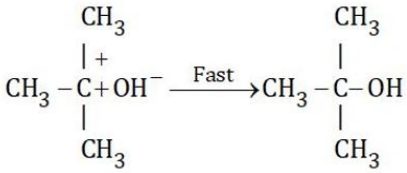
$CH_{33}CCl$

Solution:

S_N1 reaction is favoured by stability of carbocation i.e. tertiary > secondary > primary.



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Q.14 Answer:

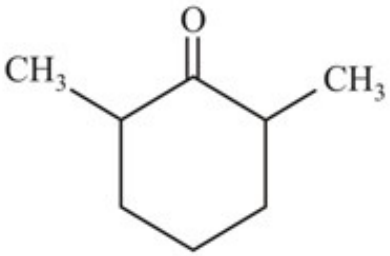
(III), (I)

Solution:

(I) 1° **allylic halide have less steric hindrance, so it is more reactive for** S_N2 **reaction.**

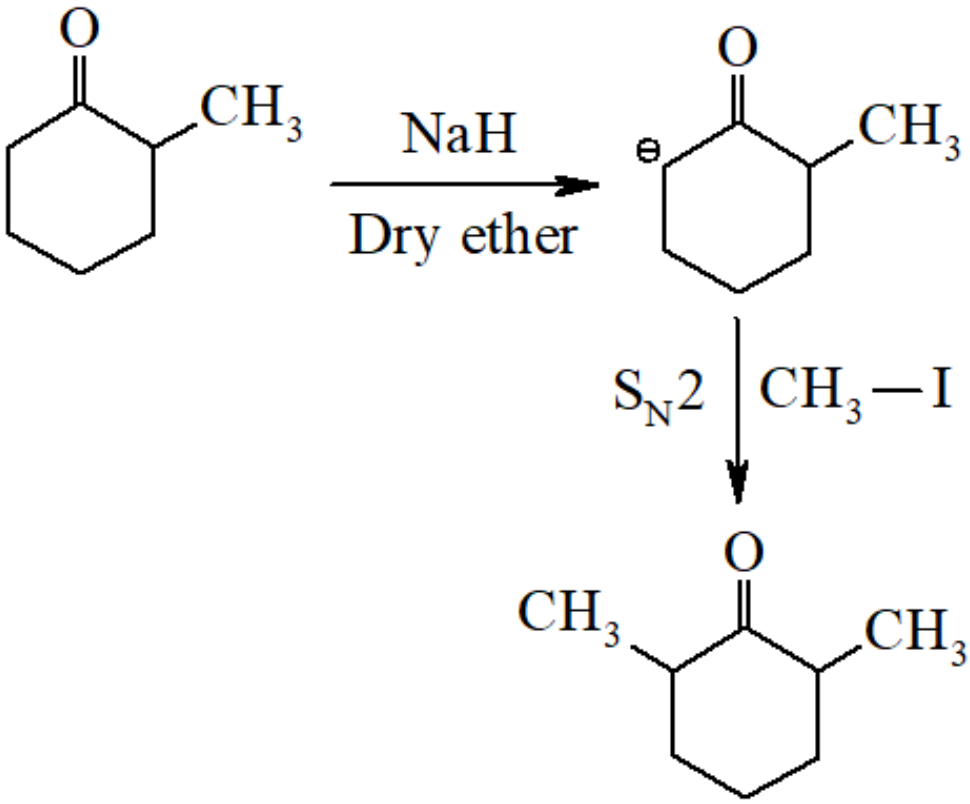
(III) 3° **allylic halide gives most stable tertiary allylic carbonium ion so most reactive towards** SN^1 **reaction.**

Q.15 Answer:



Solution:

It involves alkylation through carbanion mechanism



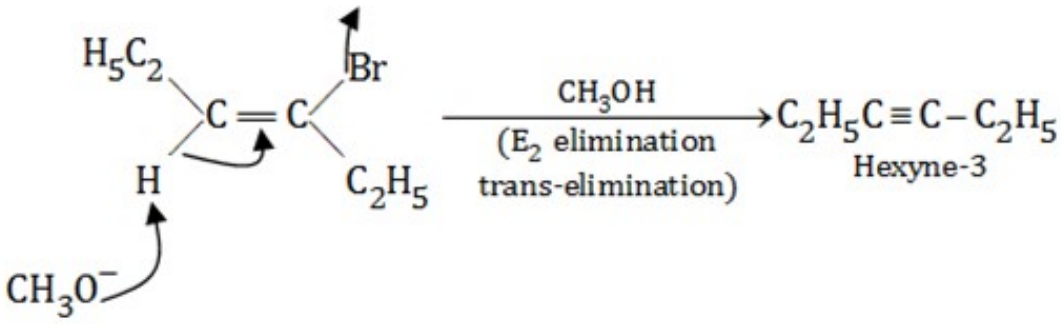
Q.16 Answer:

3-hexyne

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Solution:

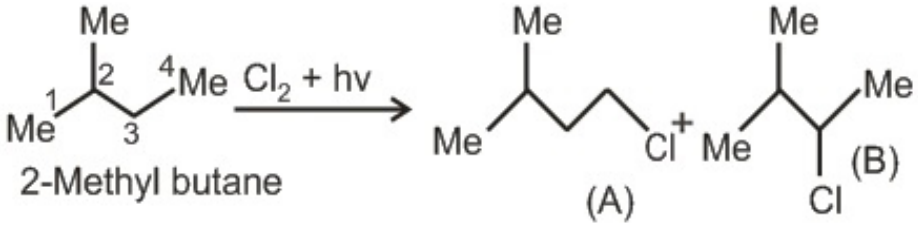
(Z)-3-bromo-3-hexene **when treated with** CH_3O^- **in** CH_3OH **gives** hex-3-yne**.**

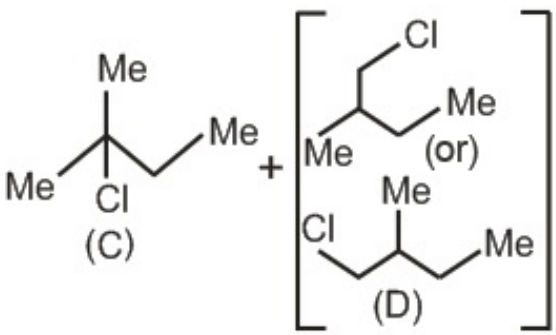


Q.17 Answer:

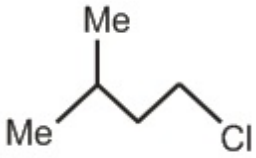
14%, 35%, 23%, 28%

Solution:

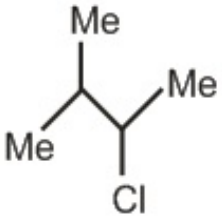
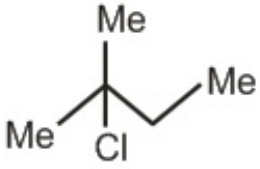
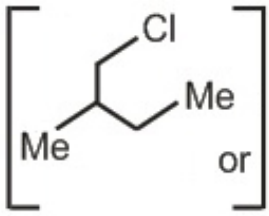
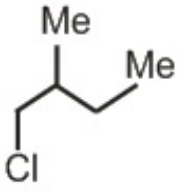




There are four different (A, B, C and D) monochloro products.

		Relative amount
(A) 	**(A) is obtained from the reaction of three equivalent 1° H atoms.**	**3 x 1 = 3.0**

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			Relative amount
(B)		**(B) is obtained from the reaction of two equivalent 2° H atoms.**	**2 x 3.8 = 7.6**
(C)		**(C) is obtained from the reaction of one equivalent 3° H atoms.**	**1 x 5 = 5.0**
(D)	 	**(D) is obtained from the reaction of six equivalent 1° H atoms.**	**6 x 1 = 6.0**
			Total = 21.6

Therefore, percentages of the A, B, C and D monochloro products are as follows :

$$A = \frac{3}{21.6} \times 100 = 13.9 \approx 14\%$$

$$B = \frac{7.6}{21.6} \times 100 = 35.18 \approx 35\%$$

$$C = \frac{5.0}{21.6} \times 100 = 23.15 \approx 23\%$$

$$D = \frac{6.0}{21.6} \times 100 = 27.8 \approx 28\%$$

(A) is obtained from the reaction of three equivalent 1°H atoms.

(B) is obtained from the reaction of two equivalent 2°H atoms.

(C) is obtained from the reaction of one equivalent 3°H atoms.

In the question: mention these 4 points A, B, C, D

(D) is obtained from the reaction of six equivalent 1°H atoms.

Q.18 Answer:

b and d

Solution:

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Vinyl halides and Aryl halides do not give Friedel Craft Reaction due to partial double bond character which is reso stabilised .

Q.19 Answer:

I^+

Solution:

In the presence of oxidising agents such as HNO_3 direct, iodination of benzene is possible. The electrophiles formed in this case is I^+ .

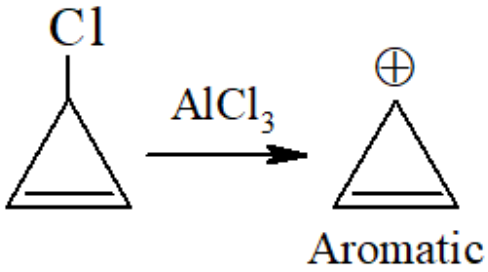
$$H^+ + HNO_3 + \frac{1}{2}I_2 \rightarrow \underset{\text{Iodine cation}}{I^+} + NO_2 + H_2O$$

Q.20 Answer:



$AlCl_3$
$\rightarrow$

Solution:



$AlCl_3$**helps in formation of carbocation.**

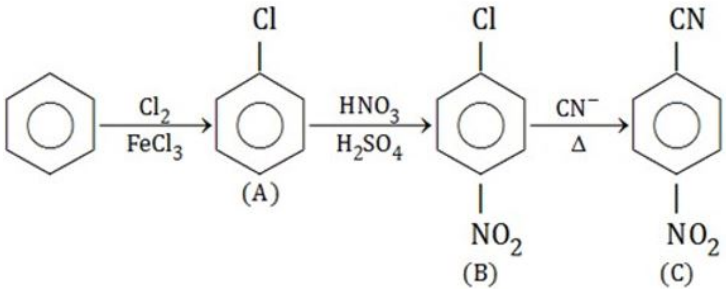
Q.21 Answer:



Solution:

The reaction sequence can be given as follows

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Q.22 Answer:



is more reactive than



Solution:

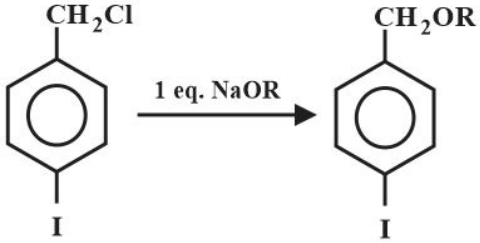
The electronegativity order of hybridized carbons is $sp > sp^2 > sp^3$. So in an allylic halide, such as $CH_2 = CH - CH_2 - I$, the alpha carbon is much more partially positive than the alpha carbon in $CH_3 - CH_2 - CH_2I$. Therefore, the attacking nucleophile will be attracted to the alpha carbon more in the allylic halide compound than in the primary halide. Hence, allylic halides tend to follow S_N2 pathways more than primary halides.

Q.23 Answer:

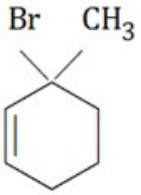


Solution:

It is a case of nucleophilic substitution:



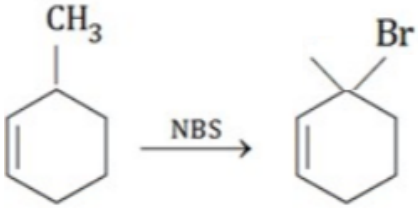
Q.24 Answer:



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Solution:

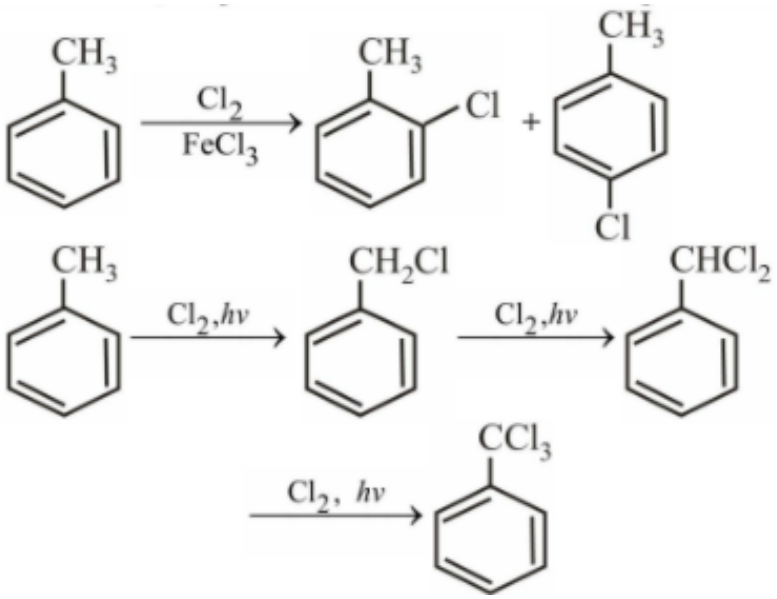
In the mono bromination of cycloalkene with NBS the bromination takes place at the allylic position.



Q.25 Answer:

X = O− and *p*− Chlorotoluene, Y = Trichloromethylbenzene

Solution:



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